

AMENDMENT NO. 3 MAY 2021

TO

IS 3431 : 1982 SPECIFICATION FOR STEEL FOR THE MANUFACTURE OF VOLUTE, HELICAL AND LAMINATED SPRINGS FOR AUTOMOTIVE SUSPENSION

(*Second Revision*)

[Page 3, Foreword **0.2** (iv)] — Substitute 'IS 8910 : 2010/ISO 404 : 1992 General technical delivery requirements for steel and steel products (*first revision*)' for 'IS : 8910-1978*'.

(Page 3, Foreword **0.3**, line 2) — Substitute 'DIN 17221-1972' for 'DIN 1722-1972'.

(Page 3, footnote*) — Delete.

(Page 4, clause **2**) — Insert the following new clause after clause **1.1** and renumber the subsequent clauses:

‘2 REFERENCES

The following Indian Standards contain provisions, which through reference in text, constitute provisions of this standard. At the time of publications the editions indicated were valid. All standards are subjected to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below:

<i>IS No</i>	<i>Title</i>
228 (various parts)	Method of chemical analysis of steel
1500 (Part 1) : 2019/ ISO 6506-1 : 2014	Metallic materials — Brinell hardness test: Part 1 Test method (<i>fifth revision</i>)

Price Group 1

<i>IS No</i>	<i>Title</i>
1586 (Part 1) : 2018/ ISO 6508-1 : 2016	Metallic materials — Rockwell hardness test: Part 1 Test method (<i>fifth revision</i>)
1608 (Part 1) : 2018/ ISO 6892-1 : 2016	Metallic materials — Tensile testing: Part 1 method of test at room temperature (<i>fourth revision</i>)
2049 : 1978	Colour code for the identification of wrought steels for general engineering purposes (<i>first revision</i>)
8910 : 2004/ ISO 404 : 1992	General technical delivery requirements for steel and steel products (<i>first revision</i>)
4163 : 2004/ ISO 4967 : 1998	Steel — Determination of content of nonmetallic inclusions Micrographic method using standard diagrams (<i>third revision</i>)

(Page 4, clause **3.1**) — Substitute ‘IS 8910’ for ‘IS : 8910-1978*’.

(Page 4, footnote*) — Delete.

(Page 4, clause **4.1**) — Substitute the following for the existing:

‘**4.1** The method of manufacture of steel shall be left to the discretion of the manufacture or as agreed to between the manufacture and purchaser.’

(Page 6, clause **6.1**) — Substitute the following for the existing:

‘**6.1** Steel shall be supplied in as-rolled condition.’

(Page 6, clause **6.1.1**) — Substitute the following for the existing:

‘**6.1.1** The steel supplied in as-rolled condition may also be supplied in one of the modes of delivery given in Table 2. In such case, the steel supplied shall be guaranteed for the properties given against different modes of delivery. The guaranteed properties are required to be met for corresponding mode of delivery as given in Table 2.’

[Page 7, clause **7.1.1** (see also Amendment No.1)] — Insert the following new clause:

‘7.1.2 The specimen used for determining through hardenability must be of a length not less than $2d$ (d is the diameter or thickness). The test piece shall be hardened in accordance with the specifications of Table 3 and then be cut into two pieces transverse to its longitudinal axis. The hardness (HRC) shall be measured according to IS 1586 (Part 1) in the middle of the cross section of the cut piece.’

(Page 7, clause 8.2) — Insert the following at the end:

‘8.2 The values shall be determined on pieces, which are to be carefully heated and held at the temperature 640°C to 680°C, for lengths of time depending on the cross sectional area and shall be as agreed to between the purchaser and the manufacturer.’

(Page 8, clause 8.3) — Insert the following new clause:

‘8.3.1 The hardness shall be tested in accordance with IS 1500 (Part 1) while tensile properties shall be tested as per IS 1608 (Part 1).’

(Page 9, clause 10.4) — Substitute ‘IS 4163’ for ‘IS : 4163-1982*’.

(Page 9, footnote*) — Delete.

(Page 11, clause 15.2.1) — Substitute ‘IS 2049’ for ‘IS : 2049-1978*’.

(Page 11, footnote*) — Delete.

(Page 11, clause 15.3) — Substitute the following for the existing:

‘15.3 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.’

(Page 11, clause 15.4) — Delete.